MIT 18.06SC - 2.7

Date ______ 20___

Problem 20.1

$$C = \begin{bmatrix} 6 & -3 & 0 \\ 3 & 1 & -1 \\ -6 & 2 & 1 \end{bmatrix}$$

$$AC^T = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 2 & 2 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 6 & 3 & -6 \\ -3 & 1 & 2 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

So $\det(A) = 3$

Changing $a_{13} = 4$ to 100 does not change $\det(A)$ because $C_{13} = 0$.

~~Problem 20.2~~

$$\cancel{\begin{bmatrix} \sin\theta & \cancel{\phi} & \rho\cos\phi\cancel{\cos\theta} & -\rho\sin\phi\sin\theta \\ & & & \end{bmatrix}}$$

Problem 20.2

$$\begin{bmatrix} \sin\phi\cos\theta & \rho\cos\theta\cos\phi & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \\ \cos\phi & -\rho\sin\phi & 0 \end{bmatrix}$$

$$\det = \sin\phi\cos\theta\,(\rho\cos\phi\sin\theta \times 0 - \rho\sin\phi\cos\theta \times -\rho\sin\phi)$$
$$+ \rho\cos\theta\cos\phi\,(+\rho\sin\phi\cos\theta\cos\phi)$$
$$- \rho\sin\phi\sin\theta\,(-\sin\phi\sin\theta\rho\sin\phi - \rho\cos\phi\sin\theta\cos\phi)$$

$$= \cancel{\rho}\sin\phi\cos\theta\,(\rho^2\sin^2\phi\cos\theta)$$
$$+ \rho\cos\theta\cos\phi\,(\rho\sin\phi\cos\theta\cos\phi)$$
$$- \rho\sin\phi\sin\theta\,(-\sin^2\phi\sin\theta\rho \cancel{\sin\phi} - \rho\cos^2\phi\sin\theta)$$

$$\cancel{= \rho^2\sin^3\phi\cos^2\theta + \rho^2\sin\phi\cos^2\theta\cos^2\phi}$$
$$\cancel{+ \rho^2\sin^3\phi\sin^2\theta + \rho^2\sin\phi\cos^2\phi\sin^2\theta}$$

$$\cancel{= \rho^2\sin^3\phi\cos^2\theta + \rho^2\sin\phi\cos^2\theta\cos^2\phi}$$
$$\cancel{+}$$

$$= \sin\phi\cos\theta\,(\rho^2\sin^2\phi\cos\theta)$$
$$+ \rho\cos\theta\cos\phi\,(\rho\sin\phi\cos\theta\cos\phi)$$
$$- \rho\sin\phi\sin\theta\,(-\rho\sin\theta\,(\sin^2\phi + \cos^2\phi))$$

$$= \sin\phi\cos\theta\,(\rho^2\sin^2\phi\cos\theta)$$
$$+ \rho\cos\theta\cos\phi\,(\rho\sin\phi\cos\theta\cos\phi)$$
$$+ \rho^2\sin\phi\sin^2\theta$$

$$= \rho^2\sin\phi\,(\sin^2\phi\cos^2\theta + \cos^2\theta\cos^2\phi + \sin^2\theta)$$

$$= \rho^2\sin\phi\,(\cos^2\theta + \sin^2\theta)$$

$$= \rho^2\sin\phi$$